PORTFOLIO PROJECT

I was tired of practice tutorials.



I wanted to do something closer to real analyst work—so I investigated a deliberately messy customer dataset.

THE BRIEF

Assess the data before cleaning it.

I used Python and pandas to identify issues, quantify their impact, document assumptions, and recommend next steps.

1,000 rows | 8 columns | 5 key issue areas

A DATA QUALITY ASSESSMENT

Not another tutorial. A report with evidence and decisions.

THE SNAPSHOT / 02

What was wrong with the data?

The dataset contains 1,000 customer records across eight columns. Every column had at least one issue to investigate.

54

signup dates missing or placeholder

5.4% of all rows

62

spend values missing or placeholder

6.2% of all rows

63

malformed email addresses

6.3% of all rows

43

extra records beyond one per ID

4.3% of all rows

THE EMAIL BREAKDOWN



906 valid (90.6%) **63 malformed** **31 missing**

Valid means the value matched the report's email pattern.

Format problems add up.

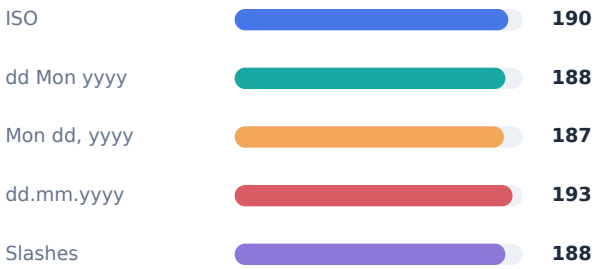
Several fields contain usable information—but not in a consistent, analysis-ready form.

SIGNUP DATES

5 formats

756 / 1,000

75.6% are not in ISO format



SPEND VALUES

408 values include a $ or USD prefix

205 with $ • 203 with USD • 40.8% of all rows

62 missing / placeholder values; stored as text

EMAILS

906 valid • 63 malformed • 31 missing



Email validation used a simple pattern check; “valid” here means pattern-matched, not verified deliverability.

Duplicates need judgment.

Counting repeated IDs is not the same as counting exact duplicate rows.

1

exact duplicate row

0.1% of all rows

42

distinct IDs appear more than once

count of customer IDs

85

rows in repeated-ID groups

8.5% of all rows

43

extra records beyond one per ID

4.3% of all rows

CATEGORY CONSISTENCY

Country spellings	14 → 5 countries	
Segment variants	15 → 4 categories	
Active-status variants	10 -> 2 values	

ID formats also differ: 720 use a 4-digit pattern; 280 use an 8-digit pattern.

The next step isn't always code.

Some changes are straightforward. Others need a decision from the data owner before remediation.

STANDARDIZE

Automate with agreed rules

- Parse dates to one agreed format
- Convert spend to numeric values
- Normalize category casing and spaces
- Validate new records at data entry

CONFIRM FIRST

Don't guess at business meaning

- Which repeated customer record is correct?
- What is the intended date order?
- Are $ and USD the same currency here?
- Are high spend values genuine or errors?

SCOPE OF THIS PROJECT

I deliberately stopped at assessment and recommendations. Cleaning should follow agreed rules, preserve the raw file, and record every change.

Nasreen Khalid • Data Quality Assessment

Python (pandas) | Dirty_customers.csv